



# FixSA Voice

## Speak. Track. Prove.

A voice-first civic issue reporting agent with resident-verified outcomes

# Track. Fix.

g agent, powered by AssemblyAI.



# Residents tell stories. Service fields

Forms push the translation burden onto the person reporting the problem. Missing context and duplicate reports slow triage, while status remains opaque.

FixSA listens first, then structures only what the workflow needs.

SUPPORTED CATEGORIES

### Everyday issues, structured for action.

The demo routes seven common service-delivery categories using a consistent data contract.



Water leaks



Potholes



Electricity faults



Sewer overflows



Broken streetlights



Illegal dumping



Damaged public assets



More via configurable routing

FOR RESIDENTS

### Less friction. More clarity.

Accessible voice and keyboard paths, plain-language prompts, a visible transcript, and a reference you can return to.

 **Your words remain yours**  
Review and edit every extracted field before anything is created. Audio is not retained in demo mode.

FOR OPERATIONS TEAMS

### Cleaner intake. Faster triage.

Structured categories, SLA rules, duplicate evidence, severity signals, and an audit trail reduce manual cleanup.

 **Built for authorised integration**  
The demo shows the workflow; a municipality or utility would still need to approve routing, identity, retention, and dispatch.

THE REASON TO WIN

## Proof of Fix.

# One conversation becomes a trackable work order

 **FixSA Voice**  
SPEAK. TRACK. FIX.

ReportTrackJudge demoOperationsAboutReport an issue

RESIDENT REPORTING

## Start with your voice.

FixSA Voice listens in real time, asks only for missing details, checks nearby reports, and waits for your confirmation.

1 Speak2 Review3 Confirm4 Track

● Deterministic demoEnglish · South African accents

### Tell us what happened.

Speak naturally. Include where it is, how long it has been happening, and any danger.



00:01  
Transcript ready

BEFORE YOU BEGIN

Demo modeReal API

ⓘ **Privacy at the microphone**  
Live audio is used only to transcribe and respond.  
FixSA does not store recordings in demo mode.

☒ **I consent to live transcription**  
I understand this is a synthetic hackathon demo.

OPTIONAL LOCATION

## Speak naturally

FixSA collects the missing location, duration and hazard detail during the conversation.

## Review before action

The resident sees the structured record and confirms the complete read-back before creation or merge.

# AssemblyAI is the interaction layer

Realtime transcript events keep the conversation visible. Spoken replies handle turn-taking. JSON-schema tool calls move validated data into the civic workflow.

## The permanent key never reaches the browser

Laravel exchanges the server-only credential for a short-lived Voice Agent token. The browser receives only the temporary session token.

**HACKATHON DEMO** Synthetic data · No report is sent to a real municipality

Report Track Judge demo Operations

1 Speak — 2 Review — 3 Confirm — 4 Track

**stay in control.**

ed fields, understand any likely duplicate, and confirm the full  
re a synthetic work order can be created or merged.

**every detail**

Severity: High

Landmark: Local clinic

People affected

**DUPLICATE CHECK**

**A nearby report may match**

**FSA-2026-1842**  
0 m away · open report

- Same category: Water leak
- Approximately 0 m away
- Overlapping description and hazard language
- Existing report is in progress

**Merge evidence** **Keep separate**

# Nine tools share one guarded contract

## CLASSIFY

classify\_service\_issue  
search\_nearby\_reports

## CONFIRM

confirm\_report\_details  
attach\_evidence

## ACT

create\_service\_request  
merge\_with\_existing\_report

## FOLLOW THROUGH

get\_report\_status  
verify\_resolution  
update\_report\_status

**Create and merge fail  
closed until the resident  
confirms the full read-  
back.**

# Proof of Fix closes the accountability loop

A status change to resolved is only a claim. The resident confirms the repair, reports a partial result, or disputes the closure.



## What the synthetic crew says

Demo repair completed; awaiting resident confirmation.



RESIDENT EVIDENCE RECORDED

## Closure challenged. Work order reopened.

"One streetlight is working, but the second light is still off."

The public timeline and operator audit trail now show the resident's outcome without retaining their audio.

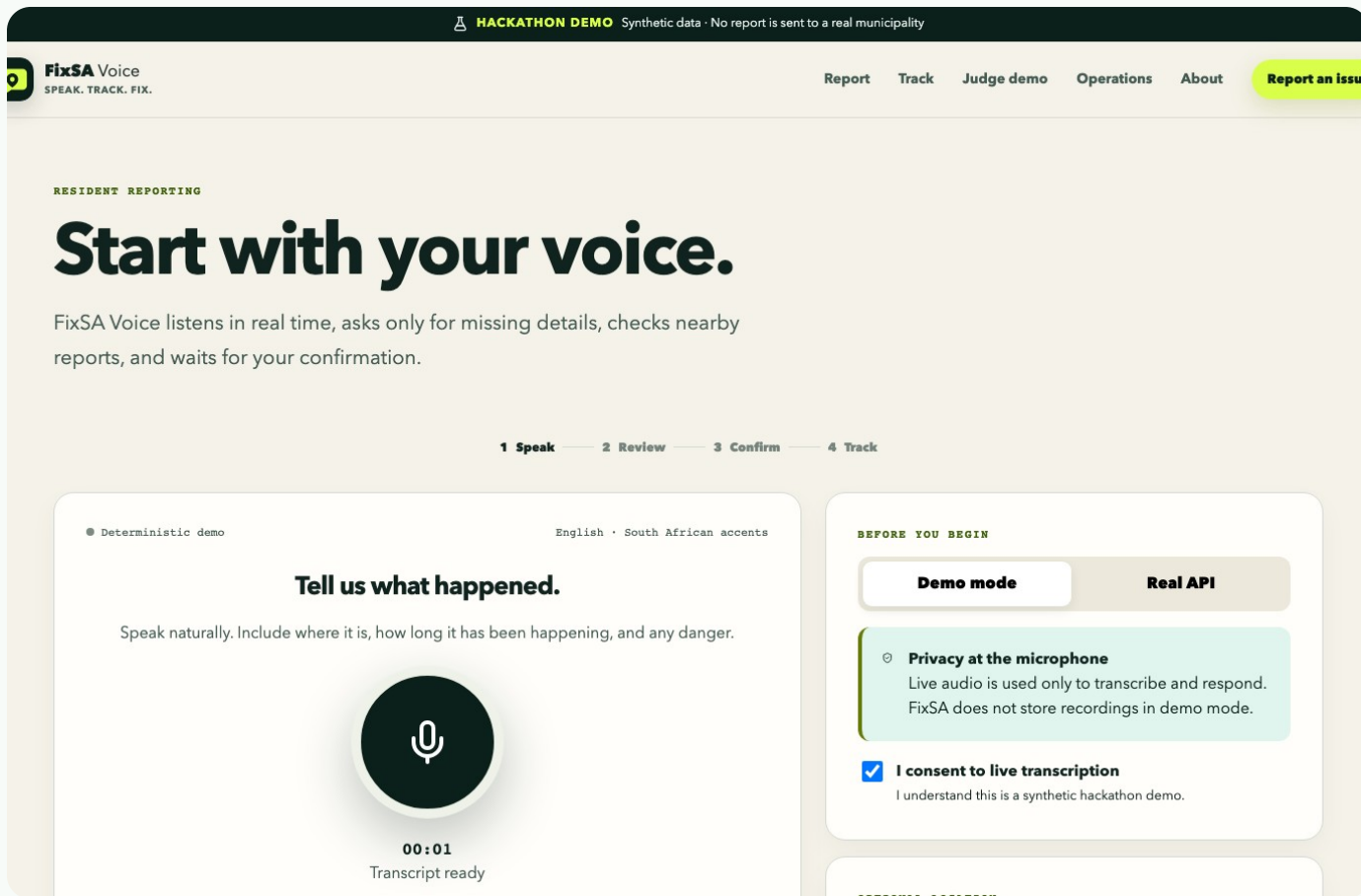
[See public proof](#)

[See operator audit](#)

## Partial or failed work reopens automatically

The public timeline and operator audit both record the resident outcome without retaining audio.

# Safety and privacy are workflow states



## Consent before capture

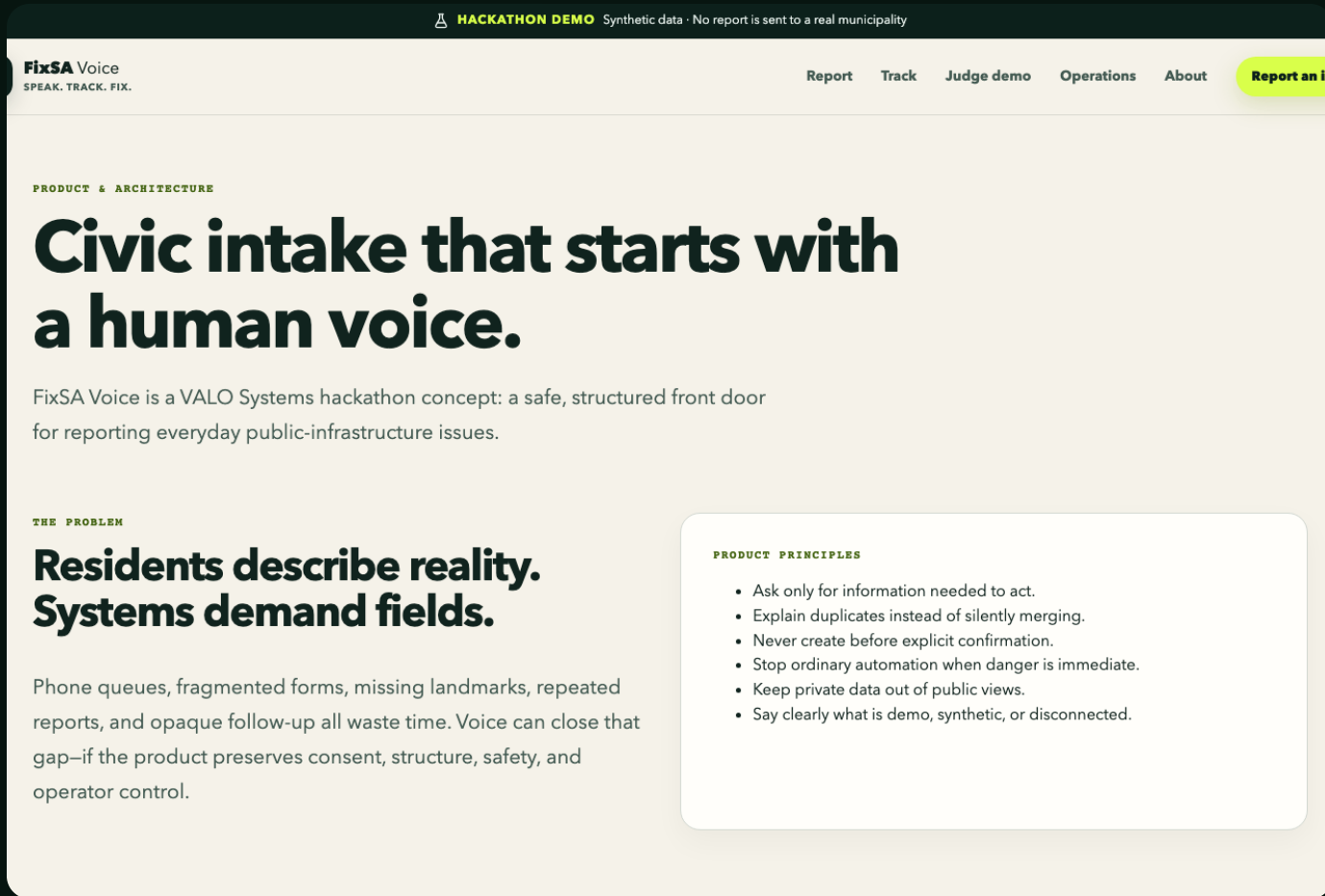
Audio stays ephemeral by default. Public views use allowlisted fields and rounded location coordinates.

## Danger stops ordinary automation

FixSA tells the resident to move to safety and contact the appropriate official service. It never fabricates an emergency number.



# Reliable demo path, production-ready data path



## Static Next.js frontend

cPanel serves the prebuilt interface over HTTPS.

## Laravel 13 and MariaDB

ULIDs, indexed public references, transactions, idempotency and immutable audit events support clean handoff to authorised systems.

## Server-only AssemblyAI key

# What works now, what remains synthetic

**BEFORE YOU BEGIN**

**Demo mode** **Real API**

 **Privacy at the microphone**  
Live audio is used only to transcribe and respond. FixSA does not store recordings in demo mode.

☐ **I consent to live transcription**  
I understand this is a synthetic hackathon demo.

**OPTIONAL LOCATION**  
**Help place the issue**



**Street, area, or landmark**

## Working now

Real AssemblyAI Voice  
Agent token flow  
Validated civic tools  
Laravel and MariaDB  
persistence  
Hosted responsive  
journeys

## Synthetic by design

People, locations and  
metrics  
Operator preview actions  
Municipal dispatch and  
integrations

**Next: authorised work-order integration, verified identity, geocoding and evidence-led language expansion.**

# Judging evidence is visible and testable

| Technology                                                                         | Originality                                                            | Trust                                                                | Delivery                                               |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------|
| Real Voice Agent token exchange<br>Nine structured tools<br>Server-only credential | Explainable deduplication<br>Resident Proof of Fix<br>Automatic reopen | Consent-first voice<br>Public/private split<br>Immediate-danger stop | 24 unit tests<br>13 API tests<br>40 local browser runs |

**Hosted acceptance confirms HTTPS, MariaDB health, CORS, redacted public tracking, 401 operator protection and temporary Voice Agent tokens.**

# Speak. Track. Prove.

**Try the resident Proof of Fix journey**

[fixsa.valosystems.co.za/verify?ref=FSA-2026-1811](https://fixsa.valosystems.co.za/verify?ref=FSA-2026-1811)

[github.com/SibusisoMashita/fixsa-voice](https://github.com/SibusisoMashita/fixsa-voice)

A VALO Systems concept / Synthetic hackathon data

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